NoteWise AI by Deep Azure

Name of Institute: **IIT, Guwahati** Team Name: **Deep Azure**

Challenge Area: **AI Study Buddy** (Track 3) Registered Email ID of your Team Lead: **priyanshurs@iitg.ac.in**

Project Title: NoteWise AI

The Team: I have a huge team of one person , myself (haha). As soon as I discovered about AI Unlocked, I decided to prove that one person if determined can create complex solutions given a problem.

As they say, down to Gehenna or up to the throne, he travels the fastest who travels alone.

The Concept: Students across schools and colleges rely heavily on handwritten notes and AI, what a coincidence. These notes are often unstructured and incomplete.
NoteWise AI transforms messy handwritten notes into a structured, interactive study companion. Using multimodal AI approach, the application:

- Reads handwritten notes from images
- Extracts and structures key topics and concepts
- Builds a knowledge base
- Explains complex concepts in simple language
- Generates adaptive quizzes
- Speaks explanations
- Supports both English and Indian languages for inclusive learning

My solution leverages OCR, LLM's, RAG, and speech technologies to create a personalized AI Study Buddy that makes revision engaging and effective.

Target Audience or Market

Primary Audience

- High school and university students (ages 15–24)
- Competitive exam aspirants (JEE,NEET,UPSC)
- Students in India who rely heavily on handwritten notes and AI

Geography

India with scalability to other multilingual regions.

Platforms

- Web application
- Future expansion: Mobile app

Market Size: According to the **All India Survey on Higher Education (AISHE) 2021-22**, India has over **43 million students enrolled in higher education**. Also, India has over **250 million school students** (Ministry of Education, Government of India).
Even if we initially target 1% of students, there are over **2 million potential users**.
Post COVID there has been a huge adoption of online learning and post AI boom, students have been continuously using it to leverage it for their studies, but no unified platforms exist yet

Personas:

Persona 1: Arjun Verma

- Age: 17
- Location: Lucknow, India
- Preparing for: JEE
- Pain Points:
    - Revising large volumes of material
    - Difficulty identifying weak topics
- Goals:
    - Targeted practice
    - Concept-based explanations
    - Audio learning during commute

Arjun uses NoteWise AI to identify weak subtopics and listens to spoken explanations while traveling. He can upload images of his notes and asks the AI to explain "S-Block elements in simple terms" and quiz him in Hindi to test him.

How it works: (*subject to change)

Step 1: Image Input & OCR: Users upload images of handwritten notes. We use **GPT-4o with Vision (via Azure OpenAI)** to extract structured text from handwritten content.
Step 2: Cleaning & Structuring
The extracted text is processed to:

- Identify key topics and subtopics
- Break content into semantic chunks
- Structure the notes logically

Step 3: Embedding & Vector Database (RAG)

Each chunk is embedded and stored in a vector database with metadata (topic, subject, page number). This enables efficient semantic retrieval and reduces token costs when querying.
Step 4: Personalized Tutor Engine
When a student asks a question:

- The query is embedded

- Relevant note chunks are retrieved
- The AI generates explanations tailored to the student's level

Step 5: Adaptive Quiz Generation
The system generates quizzes based on:

- Extracted topics
- Student interaction history
- Difficulty progression

Step 6: Multilingual & Speech Integration
Using **Azure AI Speech Services**:

- Explanations are converted to speech
- Students can hear content in English or an Indian language
- Future scope: Speech-to-text for voice-based questioning

Architecture Diagram (Text Representation)

User input image→ GPT-4o Vision text extraction → Structuring Layer → Embeddings → Vector DB (RAG) → Tutor/Quiz LLM → (TTS via Azure Speech) → Interactive UI

Microsoft Azure makes this feasible by providing:

- Unified AI ecosystem
- Seamless integration between vision, language, and speech
- Scalable cloud infrastructure
- Secure and reliable deployment environment

Core Technologies: Azure OpenAI Service (GPT-4o with Vision), Azure OpenAI Embeddings, Azure AI Speech Services, Qdrant Vector Database, Web frontend (React /Next.js)**, Python** Backend API, **Docker** for deployment.
Key concepts: Retrieval-Augmented Generation (RAG)**,** Multimodal AI (Vision+Text+Speech)**,** Adaptive Quiz Engine**,** Metadata-based semantic retrieval

The Business Plan

Monetization Strategy

Freemium model:

- Free tier: Limited uploads per month, Basic explanations
- Premium subscription: Unlimited uploads, Advanced adaptive quizzes, Multilingual voice support, Performance analytics dashboard

Future Expansion

- Institutional licensing for schools and colleges
- AI-generated revision summaries for exam prep
- Competitive exam specialized versions
- When users get accustomed, we can raise the premium subscription prices

Additional Information:

Future roadmap:

- Mobile app release
- Performance tracking dashboard
- Weak-topic heatmaps
- Gamified revision streaks

NoteWise AI reimagines studying as a conversation, transforming static notes into an intelligent tutor.